

Online Federated Learning on Distributed Unknown Data Using UAVs

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Abstract—Along with the advance of low-altitude economy, a variety of applications based on Unmanned Aerial Vehicles (UAVs) have been developed to accomplish diverse tasks. In this paper, we focus on the scenario of multiple UAVs performing Federated Learning (FL) tasks. Specifically, a group of UAVs is scheduled to repeatedly visit some Points of Interest (PoIs), collect the data produced by these PoIs, and jointly train a machine learning model based on the collected data. The most challenging issue is how to schedule UAVs to collect data so as to optimize the generalization and convergence of model training under the case that the distributions of the data produced by PoIs have not been known in advance. To address this issue, we propose a novel framework for online FL on distributed unknown data, named OFL-UD², which is dedicated to online decision-making for UAVs to optimize model training performance. Concretely, we formulate the optimization problem while considering the convergence and quality of trained models as well as energy constraints. Then, we define a utility metric for the data quality of different PoIs and conduct a rigorous convergence analysis for OFL-UD². Based on the analysis results, we design a two-stage algorithm to determine the scheduling of UAVs. Extensive simulations demonstrate that OFL-UD² can improve model accuracy and speed up running time compared to existing benchmarks significantly.

Index Terms—Federated Learning, Unmanned Aerial Vehicles, Gaussian Process, Combinatorial Multi-Armed Bandit

I. INTRODUCTION

With the advancement of smart city, some infrastructures together with policies are being built to support the development of low-altitude economy, where Unmanned Aerial Vehicles (UAVs) are organized to provide diverse services such as logistics and delivery, traffic offloading, and environment monitoring [1]–[3]. Recently, many efforts have been devoted to designing efficient and intelligent UAVs application systems serving for low-altitude economy [4]. Since Federated Learning (FL) [5]–[9] can coordinate multiple clients to jointly train a model without aggregating their local datasets, it has become one of the most important means to support UAVs in fulfilling some machine learning tasks for low-altitude economy, attracting wide attention from academia and industry.

Consider a typical UAVs application scenario, where a requester rents a group of UAVs from a service provider to conduct an FL task by visiting some Points of Interest (PoIs), collecting data from these PoIs, and jointly training a machine learning model based on the collected data. For instance, a carbon emission manager uses UAVs to collect air quality data

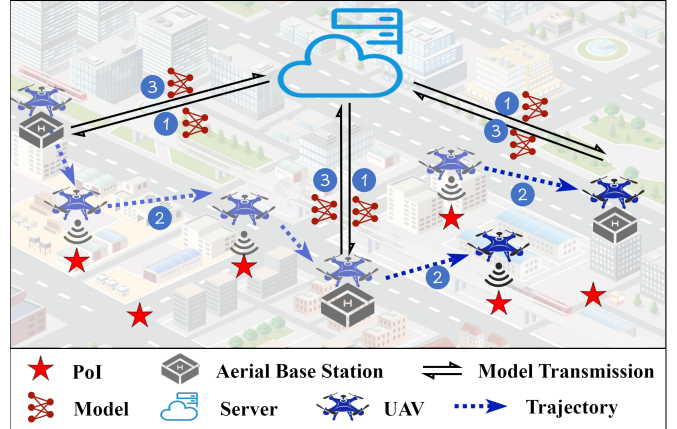


Fig. 1: The Data-driven FL system. ① The server broadcasts the model to all UAVs. ② Multiple UAVs visit PoIs, conduct data collection, and perform local model updates using collected data. ③ All UAVs upload the final local models to the server to be aggregated.

from some PoIs and applies FL to learn the relationship between oil vehicle traffic flows and carbon emission; a requester rents UAVs to collect environment images from some PoIs and train an FL model which can infer the air quality from images; and so on. In these scenarios, PoIs continuously produce large amounts of data, while UAVs do not hold any local data by themselves due to limited storage. Consequently, UAVs have to dynamically travel among multiple PoIs to collect data (e.g., directly receiving the data cached in these PoIs via Bluetooth, taking photos or videos through the built-in cameras by themselves, etc.) for model training. After local training, UAVs also need to go back to base stations periodically and communicate with the server via these stations for updating the global model. Unlike traditional FL paradigms, all training data are separated from model trainers (i.e., UAVs), which can be seen as an independent data layer. Continuous data generation drives UAVs to frequently and dynamically visit different PoIs to collect data for training, forming a novel data-driven FL paradigm, as shown in Fig. 1. Actually, many UAV-based FL applications (e.g., traffic congestion prediction [10], air quality inference [11], large-scale community electricity demand estimation, etc.) follow this paradigm.

In the above data-driven FL paradigm, FL is embedded with

a dynamic data collection process, arising two following novel challenges. The first challenge is how to make the decision for each UAV to select appropriate PoIs so that the collected data contribute to the convergence and quality of the model training mostly. On one hand, the data distributed among more PoIs can improve the generalization and quality of the trained model. On the other hand, the heterogeneity of data collected from PoIs and the introduction of low-quality PoI data might also degrade the convergence of model training. What's more, it is hard to measure the extent of convergence degradation incurred by data heterogeneity before model training. These factors make the PoI selection become a nontrivial online decision issue. The second challenge comes from the unknown data distribution. Since UAVs do not know the distribution of data produced by different PoIs in advance, they have to learn these data distributions while making online decisions, which involves a balance issue of so-called exploration and exploitation, i.e., using the learned knowledge to make the current optimal decision or exploring the knowledge that has not been learned before. Such an issue makes the above online decision much more challenging. To the best of our knowledge, this is the first work to investigate how to schedule UAVs to select PoIs and collect unknown data for efficient model training under a novel data-driven FL paradigm, where all training data are separated from the model trainer (i.e., UAVs) as an independent data layer.

So far, many efforts have been devoted to improving the performance of model training and the model quality through selecting clients together with their data according to specific strategies [12]–[16]. For instance, Gong *et al.* [15] propose a data valuation metric based on inference accuracy, combined with known data distribution for on-device data selection. Jaemin *et al.* [14] prioritize informative data via gradient norm to accelerate global training. Ravikumar *et al.* [16] select clients with representative gradient information using submodular maximization. Unlike these works, clients (i.e., UAVs) in our FL system do not hold local data by themselves, and the data distributions are unknown in advance. UAVs have to select PoIs and explore the data distribution in each round of FL. PoI selection and learning unknown data might influence the convergence and quality of model training, and the influence cannot be expressed explicitly. Existing works cannot deal with this novel problem.

To overcome the above challenges, we propose an online FL framework for distributed unknown data, namely, OFL-UD², which can make online decisions for each UAV in unknown environments to optimize the generalization and convergence of FL model training. Specifically, we first define the data utility for each PoI by considering the generalization of trained models, and formulate the optimization problem with the convergence and energy constraints to maximize the total data utility. In order to understand the impact of PoI data on model convergence, we then conduct a thorough convergence analysis of online FL and distinguish a data quality metric of different PoIs based on the convergence upper bound. Based on the analysis results, we design a two-stage algorithm to determine

the scheduling of UAVs. In the first stage, we propose a novel PoI selection strategy within the OFL-UD² framework, which leverages Gaussian Processes (GP) and Combinatorial Multi-Armed Bandits (CMAB) to quickly capture the nature of PoI data through iterative online learning. This approach dynamically balances the exploration and exploitation of PoIs under the unknown and uncertain environment. By progressively converging to the optimal PoI selection, OFL-UD² provides a distinct advantage in handling the dynamic nature of PoI data. In the second stage, we employ a real-time trajectory control scheme for UAVs to minimize energy consumption. In a nutshell, our major contributions are summarized as follows:

- We propose a novel online data-driven FL framework for distributed unknown data, termed OFL-UD², which can effectively make online decisions for UAVs to maximize model training performance. Different from existing works, our data-driven FL framework can guarantee both the generalization and convergence of model training with distributed unknown data.
- We derive a rigorous convergence bound to transform the convergence error into quantifiable metrics, based on which our problem is decomposed into a two-stage optimization problem to be solved.
- We design a GP-CMAB-based PoI selection strategy, enabling UAVs to be scheduled to optimal PoIs for near-optimal model training performance through multi-round online learning under the unknown environment. Additionally, we derive a sub-linear regret bound.
- Extensive simulations are conducted to evaluate the performance of OFL-UD². The results demonstrate that OFL-UD² can improve the final model accuracy by 6.83% to 16.58%, with a speedup by about $1.2\times$ to $3\times$, compared to the baselines.

II. RELATED WORKS

We review the related work from the following two aspects:

Integration of FL and UAVs: Since the emergence of FL, numerous studies have combined FL with UAVs. For instance, various works in [11], [17] deployed FL to improve the efficiency of UAV applications, such as quality sensing, secure crowdsensing using blockchain, and real-time control of large UAV fleets. In contrast to these works that focused on UAV applications themselves, some other studies aimed to enhance FL applications using UAVs. For example, Pham *et al.* [18] proposed a scheme where UAVs power FL users to mitigate battery constraints in FL communication. Hou *et al.* [19] developed a covert FL architecture with UAVs emitting noise to prevent eavesdropping. Lim *et al.* [20] introduced a UAV-based collaborative learning approach for vehicular networks, while Zhang *et al.* [21] examined UAV-assisted exploration for image classification, closely aligning with our scenario. Nevertheless, they focused on reducing the communication cost under imperfect channels. In summary, these studies either treat UAVs as base stations for scheduling or assume that data is already available on the UAVs, overlooking the impact of PoI selection on FL convergence.

Data valuation in FL: Selecting informative and valuable samples for training can reduce computational and communication resource consumption. Li *et al.* [13] proposed a method to prioritize training samples from more important clients in FL by using the gradient norm upper bound. Shin *et al.* [14] evaluated sample utility based on loss, drawing a higher proportion from high-loss samples. Such importance sampling methods often require full dataset traversal and outlier removal. Gong *et al.* [15] quantified the impact of local data samples on global model performance through the inner product of the local sample gradient and global gradient. Some client selection schemes evaluate data quality from alternative perspectives. Wang *et al.* [22] developed a method prioritizing clients with higher local variance, considering both client diversity and local variance. Similar to importance sampling, Li *et al.* [23] evaluated the utility of each client based on data volume and loss. Saha *et al.* [12] proposed a client selection approach based on data volume and distribution skew. However, these works often assume that the training data are static, which cannot address our unique challenges.

III. SYSTEM MODEL AND PROBLEM STATEMENT

A. System Overview

As illustrated in Fig.1, we consider a typical data-driven FL system, where a requester rents a group of UAVs from a service provider to strategically visit some Points of Interest (PoIs). Since each PoI continuously produces large amounts of data and it is time-consuming to transmit these data, UAVs need to process data in real-time and make full use of computing resources to accomplish an FL task (e.g., traffic signal analysis) cooperatively. Let $\mathcal{N} = \{1, 2, \dots, N\}$ and $\mathcal{M} = \{1, 2, \dots, M\}$ denote the group of UAVs and M PoIs, respectively. Here, each PoI $m \in \mathcal{M}$ generates samples in the form of data streams, and these data samples follow a *unknown* distribution, denoted by q_m . For the sake of energy efficiency and communication reliability, there are several Aerial Base Stations (ABSs), denoted by $\mathcal{S} = \{1, 2, \dots, S\}$, to provide charging facilities and act as communication relays. Besides, the data-driven FL system operates in a time-slotted manner, and the whole process is divided into R rounds. For each round $r \in \{1, 2, \dots, R\}$, the workflow of our system is delineated as:

Step 1: Online Decision-Making for UAV Scheduling. According to the designed framework (i.e., Section V), the cloud-based service provider determines the scheduling of all UAVs for data collection and training, denoted by $\mathcal{A}(r) = \{A_1(r), \dots, A_n(r), \dots, A_N(r)\}$. Here, $A_n(r)$ represents an ordered set of PoIs for each UAV n , specifying the sequence in which UAV n will visit the PoIs. Let c_n be the total number of PoIs visited by UAV n . Then we denote $A_n(r) = \{m_1^n, \dots, m_i^n, \dots, m_{c_n}^n\}$, $\forall n \in \mathcal{N}$, where m_i^n is the index of the i -th visited PoI by UAV n . For simplicity, we also define $\mathcal{P}_r = \cup_{n \in \mathcal{N}} A_n(r)$ to denote the representative subset of PoIs in round r . In order to improve the data collection efficiency of UAV under limited energy, we set $|\mathcal{P}_r| = M_0$, and $M_0 > N$. The service provider then broadcasts the latest global model and decision variable $A_n(r)$ to each UAV n .

Step 2: Data Collection and Local Model Training.

Upon receiving the decision variable, each UAV n departs from the ABS and sequentially arrives at the designated PoIs according to the order specified in the moving trajectory $A_n(r)$. Subsequently, UAV n flies in a hovering mode and establishes a virtual multiple-input multiple-output link [24] to collect data from the corresponding PoIs.

After accomplishing the data collection process, each UAV n performs local training by exploiting the collected data $\mathcal{D}_r^n$. We define $l(\mathbf{w}, \xi)$ as a loss function to measure how well a machine learning model $\mathbf{w}$ performs on the data sample ξ . The local loss function for UAV n is given by:

$$F_{r,n}(\mathbf{w}; \mathcal{D}_r^n) = \frac{1}{|\mathcal{D}_r^n|} \sum_{\xi \in \mathcal{D}_r^n} l(\mathbf{w}, \xi). \quad (1)$$

In this paper, we adopt the mini-batch Stochastic Gradient Descent (SGD) algorithm for local training with $k \in 1, 2, \dots, K$ indexing the local SGD iterations. Let $\mathbf{w}_r$ and $\mathbf{w}_{r,n}$ represent the global model and the local model of UAV n , respectively. Using the latest global model parameters $\mathbf{w}_{r-1}$, we initialize $\mathbf{w}_{r,n}^0 = \mathbf{w}_{r-1}$. Each UAV then n updates its local model using the local dataset $\mathcal{D}_r^n$ for K iterations, as expressed below:

$$\mathbf{w}_{r,n}^{k+1} = \mathbf{w}_{r,n}^k - \eta_r \nabla F_{r,n}(\mathbf{w}_{r,n}^k, \xi), k = 1, 2, \dots, K. \quad (2)$$

Here, ξ denotes a random mini-batch selected from the local dataset $\mathcal{D}_r^n$, and η_r is the learning rate in round r .

Step 3: Global Model Update. To ensure UAVs have sufficient energy for continuous operation, each UAV needs to return to the nearest ABS after completing local model training. Subsequently, each UAV uploads its final local model $\mathbf{w}_{r,n}^K$ to the service provider and recharges. The service provider then aggregates the received local models to form a new global model for the next training round, i.e., $\mathbf{w}_r = \sum_{n \in \mathcal{N}} p_{r,n} \mathbf{w}_{r,n}^K$. Here, $p_{r,n}$ is the aggregation weight for UAV n in the r -th round, and there is $\sum_{n \in \mathcal{N}} p_{r,n} = 1$.

Essentially, the training rule exhibits no deviation from the local update rule of vanilla FedAvg [6]. The system iterates the above steps for R rounds. Overall, over the entire service period, we define the global loss function for UAVs as:

$$F(\mathbf{w}) \triangleq \frac{1}{R} \sum_{r=1}^R \sum_{n=1}^N p_{r,n} F_{r,n}(\mathbf{w}; \mathcal{D}_r^n). \quad (3)$$

The goal of the data-driven FL system is to obtain the optimal model parameters $\mathbf{w}^*$ so as to minimize $F(\mathbf{w})$, i.e.,

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} F(\mathbf{w}). \quad (4)$$

B. Problem Formulation

Energy efficiency is a crucial requirement for UAV-aided systems. For each round in OFL-UD² framework, the energy consumption of a UAV can be divided into two parts. The first part arises from the power consumption for flying between different PoIs and hovering to collect data, denoted as E_{travel} . The second part is the energy consumption during the learning phase for training and communication, denoted as E_{train} . The total energy consumption for UAV n in round r can be expressed as $E_n(r) = E_{travel} + E_{train}$.

Similar to [25], when a UAV is flying at a fixed speed v , the power can be given by the expression:

TABLE I: Description of major notations

Variable	Description
$\mathcal{M}, m$	the set of PoIs and the index of an PoI.
$\mathcal{N}, n$	the set of UAVs and the index of a UAV.
$\mathcal{S}, s$	the set of ABSs and the index of a ABS.
$\mathcal{R}, r$	the set of rounds and the index of a round.
$\mathcal{A}(r)$	the scheduling strategy of UAVs in round r .
$\mathcal{P}_r$	the subset of PoIs to visit in round r .
M_0	the total number of PoIs to visit in one round.
$\mathbf{w}_{r,n}$	the local model parameter on UAV n in round r .
η_r	the learning rate of local training in round r .
K, k	the total number and index of iterations in each round.
H_m, q_m	the information entropy and data distribution of PoI m .
$ D_r^m $	the data volume for PoI m in round r .
$U_m(r)$	the utility function for PoI m in round r .
$E_n(r)$	the energy consumption of UAV n in round r .
E_0	the total energy budget for all UAVs in one round.
σ_m	the gradient variance of loss function on PoI m .
G_m	the gradient bound of loss function on PoI m .
$h_m(j)$	the reward when PoI m has not been played for j rounds.

$$P_{fly} = p_1(1 + \frac{3v^2}{U_p^2}) + p_2\sqrt{(1 + \frac{v^4}{4v_0^2})^{1/2} - \frac{v^2}{2v_0^2}} + \frac{1}{2}p_3v^3, \quad (5)$$

where p_1, p_2 , and p_3 are blade profile power, induced power due to derivative drag, and aerodynamic drag constants, respectively. U_p characterizes the rotational blade tip speed, and v_0 is the average induced velocity during hover. Correspondingly, the power of the UAV while hovering to collect data is represented as $P_{hover} = p_1 + p_2$. Then, the power consumed by each UAV n in round r can be computed as: $E_{travel} = \sum_{i=1}^{c_n-1} (P_{fly} \cdot \frac{d(m_i^n, m_{i+1}^n)}{v} + P_{hover} \cdot t_c)$, where $d(m_i^n, m_{i+1}^n)$ is the distance between PoI m_i^n and m_{i+1}^n . Meanwhile, we define e_1 as the energy consumption for each local update and let e_2 be the energy consumption for communication during the global aggregation process. Accordingly, the energy consumption during the training phase is $E_{train} = K \cdot e_1 + e_2$.

Data distribution is an important characteristic of the data stream produced by each PoI. Like most existing studies [15], [26], we make the following assumptions:

Assumption 1. (Stationary data distribution). The data samples on PoI m are drawn independently from a fixed unknown distribution q_m , for all $m \in \mathcal{M}$.

To quickly select suitable PoIs for model training, we further define the data entropy and a utility function for each PoI:

Definition 1. (Data Entropy) Let $\{1, \dots, b, \dots, B\}$ be the label set of data samples for all PoIs, and we define q_m^b to represent the proportion of data with label b generated at PoI m . Therefore, the data distribution at PoI m can be simply expressed as $q_m = (q_m^1, q_m^2, \dots, q_m^B)$, and there exists $\sum_{b=1}^B q_m^b = 1$. Similar to existing studies [27], we define the data entropy of PoI m as $H_m = -\sum_{b=1}^B q_m^b \log q_m^b$.

The entropy H_m quantifies the diversity and informativeness of data samples at PoI m and is directly related to the data distribution q_m . A more uniform data distribution yields higher entropy, which in turn enhances the generalization performance of trained models. Furthermore, at a fixed entropy

level, increasing the volume of data can reduce overfitting, thereby minimizing the generalization error of the models [28]. Leveraging these insights, we define the utility function.

Definition 2. (Utility Function of PoI) Considering the positive impact of higher entropy and larger data volume on model performance, we define the utility function $U_m(r)$ for PoI m in round r as the following equation, i.e.,

$$U_m(r) \triangleq |D_r^m| \cdot H_m. \quad (6)$$

In the r -th round, UAVs fly to the PoIs to collect and process data. Due to limited resources, UAVs can only select a subset of PoIs (i.e., $\mathcal{P}_r$) for distributed processing. Therefore, the primary goal for the service provider is to address a sequential decision problem, i.e., determining the scheduling of UAVs $\mathcal{A} = \{\mathcal{A}(1), \mathcal{A}(2), \dots, \mathcal{A}(R)\}$, so as to maximize the expected cumulative utility. Additionally, multiple UAVs adopt an FL framework to process data in a distributed manner. Hence, it is necessary to ensure the convergence performance of the global model. That is, the service provider needs to restrict that the expected global loss converges to the optimal loss with an ϵ precision. Considering the limited availability of resources, we also restrict the energy consumption in each round. Therefore, this optimization problem can be formulated as:

$$\mathbf{P1}: \quad \max_{\mathcal{A}} \sum_{r=1}^R \sum_{m \in \mathcal{P}_r} |D_r^m| \cdot H_m, \quad (7)$$

$$\text{s.t.} \quad \mathbb{E}[F(\mathbf{w}_R)] - F(\mathbf{w}^*) \leq \epsilon, \quad (8)$$

$$\sum_{n \in \mathcal{N}} E_n(r) \leq E_0, \quad \forall r \in \mathcal{R}. \quad (9)$$

Here, E_0 is the threshold of the energy consumption. Eqs. (8) and (9) means the model convergence constraint and the energy budget constraint, respectively.

Solving Problem **P1**, however, is challenging in two aspects.

1) Before actually training the model, it is generally unclear how the choice of PoI will affect the final model $\mathbf{w}^R$ and the corresponding loss function $F(\mathbf{w}^R)$. There is still no theoretical understanding of the impact of different PoI data on model convergence in data-driven FL systems. Therefore, we need to derive an expression to explore the complex impact of the local data of multiple PoIs on model performance.

2) In each round, UAVs prioritize PoIs with higher utilities. However, since the data at each PoI is continuously generated and cleared, the utility functions of different PoIs dynamically change over time. Therefore, the dynamic utility function is not only unknown but also varies across multiple PoIs, making it impossible to impose specific prior assumptions on the utility function. In such an ever-changing and unknown scenario, it is more difficult to make online and sequential decisions to carefully balance exploration and exploitation.

IV. CONVERGENCE AND ANALYSIS

In this section, we address the first challenge by conducting a rigorous convergence analysis of our data-driven FL system, emphasizing how data on different PoIs influences the global model. Different from traditional analysis, we highlight the influence of data selection on convergence error and introduce a new PoI data value metric based on this analysis.

To facilitate a tractable convergence analysis, we state several assumptions regarding the loss function at each PoI.

Assumption 2. (*L-smooth*). For each PoI $m \in \mathcal{M}$ and round $r \in \mathcal{R}$, $F_{r,m}$ is *L-smooth*: for all $\mathbf{v}$ and $\mathbf{w}$, $F_{r,m}(\mathbf{v}) \leq F_{r,m}(\mathbf{w}) + (\mathbf{v} - \mathbf{w})^T \nabla F_{r,m}(\mathbf{w}) + \frac{L}{2} \|\mathbf{v} - \mathbf{w}\|_2^2$.

Assumption 3. (*Strongly-convex*). For each PoI $m \in \mathcal{M}$ and each round $r \in \mathcal{R}$, $F_{r,m}$ is ν -strongly convex: for all $\mathbf{v}$ and $\mathbf{w}$, $F_{r,m}(\mathbf{v}) \geq F_{r,m}(\mathbf{w}) + (\mathbf{v} - \mathbf{w})^T \nabla F_{r,m}(\mathbf{w}) + \frac{\nu}{2} \|\mathbf{v} - \mathbf{w}\|_2^2$.

Assumption 4. (*Unbiased gradient and bounded variance*). Let ξ_m^k be a sample selected from PoI m uniformly at random. The stochastic gradient is an unbiased estimator of the local gradient: $\mathbb{E}[\nabla F_{r,m}(\mathbf{w}_{r,m}^k, \xi_m^k)] = \nabla F_{r,m}(\mathbf{w}_{r,m}^k)$. Meanwhile, the stochastic gradient has bounded variance: $\mathbb{E}[\|\nabla F_{r,m}(\mathbf{w}_{r,m}^k, \xi_m^k) - \nabla F_{r,m}(\mathbf{w}_{r,m}^k)\|^2] \leq \sigma_m^2$, for $m \in \mathcal{M}$.

Assumption 5. (*Bounded local gradient*). The expected squared norm of stochastic gradients in each PoI is bounded. i.e., $\mathbb{E}[\|\nabla F_{r,m}(\mathbf{w}_{r,m}^k)\|^2] \leq G_m^2$, for all $m \in \mathcal{M}$.

Based on the assumptions outlined above, we present the theoretical analysis of the convergence bound.

Theorem 1. We define $\bar{\eta} = \min_r \{\eta_r\}$ and $\tilde{\eta} = \max_r \{\eta_r\}$. Let Assumptions 1 to 4 hold, and set the learning rate to satisfy $\bar{\eta} \leq \frac{2}{\mu}$. Given the PoI selection strategy $\mathcal{P} = \{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_R\}$, after R rounds training, the difference between the expected global training loss $\mathbb{E}[F(\mathbf{w}_R)]$ and the optimal value $F(\mathbf{w}^*)$ satisfies

$$\mathbb{E}[F(\mathbf{w}_R)] - F(\mathbf{w}^*) \leq \frac{L}{2} \left(1 - \frac{\nu \bar{\eta}}{2}\right)^R \|\mathbf{w}_0 - \mathbf{w}^*\|^2 + \frac{L\beta_1}{2M_0\nu} \sum_{r=1}^R \sum_{m \in \mathcal{P}_r} [G_m^2 + \sigma_m^2], \quad (10)$$

where $\beta_1 = \tilde{\eta}[\nu \tilde{\eta} K^2 + 2(K-1)^2 + 1]$ is a function of the training process with respect to the hyperparameters.

Theorem 1 clearly presents the relationship between various factors and global loss in our data-driven FL system. It can be observed that the term $\|\mathbf{w}_0 - \mathbf{w}^*\|^2$ representing the distance between the initial model parameters and the optimal model parameters, can not be tunable. Let $\Phi_m = G_m^2 + \sigma_m^2$ and the $\sum_{m \in \mathcal{P}_r} \Phi_m$ term depends on the PoI selection in round r . Thus Theorem 1 quantifies the impact of PoI selection on the convergence bound. In general, tighter convergence upper bounds correspond to better optimization results. Therefore, in order to obtain a better global model within limited rounds, one should prefer to choose PoIs with smaller Φ_m .

Theorem 1 inspires us to use Φ_m as the data valuation metric of PoI. This metric consists of the gradient variance σ_m and gradient upper bound G_m at each PoI m . Gradient variance, resulting from the random sample selection, reflects the stability of optimization in mini-batch gradient updates [29]. This is consistent with our understanding that greater gradient variance and higher upper bounds indicate lower data quality, leading to increased convergence error. To further validate the effectiveness of the data valuation metric, we conduct a series of experiments in Fig. 2. After injecting

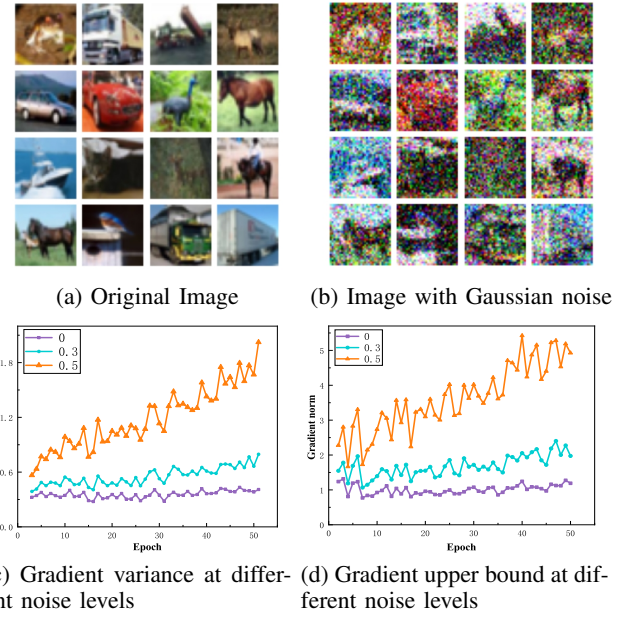


Fig. 2: Experiments for empirical observations.

the Gaussian noise into the original images, we observe the changing trends of Φ_m under different noise levels. As shown in Figs. 2c and 2d, we find that both gradient variance and gradient upper bounds increase as data noise intensifies. During the training process, the gradient variance and gradient upper bound of data with different noise levels maintain a certain stability. Therefore, Φ_m can be used as a good data quality metric for PoI m .

The uniqueness and novelty of Theorem 1 are reflected in two main aspects. First, the traditional federated analysis assumes that each client's training data is static, whereas in our data-driven FL paradigm, each UAV's training data (collected from various PoIs in each round) is dynamic. This dynamic nature complicates the convergence analysis; detailed proof is provided in the appendix. Second, existing works [30] assume uniform upper bounds on gradient variance, while PoIs generally have different upper bounds on gradient variance. More importantly, a specific correlation exists between gradient variance and the data quality of each PoI.

V. ALGORITHM DESIGN

In this section, we first transform the convergence constraint into measurable metrics using the convergence upper bound. Then, we introduce the Gaussian Process to learn dynamic reward functions on different PoIs and select the optimal PoI subset. Finally, we give a regret analysis of the algorithm, followed by the detailed framework of OFL-UD².

A. Problem Transformation

In Problem **P1**, we need to restrict that the convergence error under a fixed number of iterations is less than a specified precision ϵ . However, it is generally impossible to know how different PoI selections affect the actual error. Therefore, we leverage the convergence upper bound derived in Theorem

1 to approximate the actual error, and Problem **P1** can be reformulated into the following optimization objective:

$$\mathbf{P2:} \quad \max_{\mathcal{A}} \sum_r \sum_{m \in \mathcal{P}_r} |D_r^m| \cdot H_m, \quad (11)$$

$$\text{s.t.} \quad \sum_{r=1}^R \sum_{m \in \mathcal{P}_r} [G_m^2 + \sigma_m^2] \leq \epsilon_0, \quad (12)$$

$$\sum_{n \in \mathcal{N}} E_n(r) \leq E_0, \quad \forall r \in \mathcal{R}. \quad (13)$$

where $\epsilon_0 = \frac{2M\epsilon}{\beta_1 L} - \frac{1}{\beta_1} (1 - \frac{\nu\eta}{4})^R \|\mathbf{w}_0 - \mathbf{w}^*\|^2$.

Problem **P2** is a stochastic optimization problem with several unknown parameters: the information entropy H_m and the amount of data $|D_r^m|$ collected at each PoI m in round r . Meanwhile, the scheduling strategy $\mathcal{A}$ in Problem **P2** also needs to satisfy both convergence and energy constraints. It is evident that the optimization objective and the convergence constraint are solely related to the selection of PoIs. Therefore, we design a two-stage algorithm to tackle this problem: the first stage determines the set of selected PoIs, and the second stage assigns PoIs to UAVs based on minimal energy consumption. In the following Section V-A, we propose a GP-CMAB-based PoI selection strategy to solve the PoI selection problem. In Section V-B, we address the assignment of PoIs to UAVs and then present the overall framework for OFL-UD².

B. GP-CMAB-based PoI Selection Strategy

We first apply the Lagrange dual approach to transform the PoI selection problem (i.e., Eqs. (11) and (12)) into a min-max problem, which is shown as follows:

$$\min_{\lambda} \max_{\mathcal{P}} \mathcal{L}(\mathcal{P}, \lambda), \quad (14)$$

$$\text{s.t.} \quad \lambda \geq 0. \quad (15)$$

Here, the Lagrange dual function is given by:

$$\mathcal{L}(\mathcal{P}, \lambda) = \sum_{r=1}^R \sum_{m \in \mathcal{P}_r} (|D_r^m| \cdot H_m - \lambda(\sigma_m^2 + G_m^2)) + \lambda \epsilon_0, \quad (16)$$

where λ is the lagrange multiplier. To solve this min-max problem, we first address it by deriving the optimal PoI subset for any given λ . Since we can ignore the invariant constraint terms, we focus solely on optimizing the terms related to the PoI selection process. In other words, we regard the term $|D_r^m| \cdot H_m - \lambda(\sigma_m^2 + G_m^2)$ as the reward for selecting each PoI m . In this way, the goal of the PoI selection problem is to maximize the cumulative reward without any restrictions.

Selecting a set of PoIs with the largest reward value under the circumstance that their rewards are unknown a priori is actually an online learning and decision-making process, so we model it as a Combinatorial Multi-Armed Bandit (CMAB) problem. CMAB is a widely used reinforcement learning model for online decision-making in uncertain environments [31]. The player will pull some arms round by round according to the bandit strategy to maximize the cumulative reward. To this end, each PoI can be regarded as an arm, the service provider is seen as the player, visiting PoI m is equivalent to pulling arm m , and $|D_r^m| \cdot H_m - \lambda(\sigma_m^2 + G_m^2)$ is the corresponding reward. In particular, the reward of any arm changes at any time, regardless of whether it is played. Concretely, the reward function increases along with the growth of the data volume, and it also drops sharply after the data

Algorithm 1: GP-CMAB-based PoI Selection

Require: $\mathcal{M}$, α_r , λ , $J_{m,r-1}$, $\mathbf{Y}_{m,C}$ for all $m \in \mathcal{M}$;

Ensure : the representative subset of PoIs, $\mathcal{P}$;

```

1 for PoI  $m = 1, \dots, M$  do
2   Update the covariates  $J_{m,r}$  according to Eq. (17);
3   Calculate kernel matrix  $\mathbf{G}_C = [g_m(j_i, j_i)]_{i,l=1}^C$ ;
4   Train GP-CMAB model with  $\mathbf{G}_C$  and observations;
5   Update  $\mu_r(m)$  and  $\delta_r^2(m)$  by Eqs. (18) and (19);
6   Calculate  $\Psi_r(m)$  by Eq. (20);
7 Sort the PoIs according to the UCB value  $\Psi$ ;
8 Select the top  $M_0$  PoIs as  $\mathcal{P}_r$ ;
9 Observe the feedback for each selected PoI  $m$ ;
10 Return the set of selected PoIs in round  $r$ ,  $\mathcal{P}_r$ .
```

is collected. Since we are not clear about the dynamics of reward functions, the online decision-making process in such a dynamically changing scenario is a challenging issue. This is a typical restless bandit problem, where each arm's average reward varies over time. Although several works [32]–[34] address dynamic scenarios, these algorithms are not directly applicable to our problem for two main reasons. First, the reward distribution of an arm varies according to both the time period itself and whether the arm is selected by the strategy, while the mean reward changes irrespective of the strategy in general restless settings. Second, methods (e.g., the Whittle index [32] and upper confidence bandit [35]) estimate the reward of each arm through a specific parametric function, which is not reasonable in our heterogeneous PoI scenario. Unlike these methods, we incorporate the Gaussian Process (GP) within a Bayesian framework and propose a GP-CMAB model, which can learn dynamic reward functions for multiple PoIs and provide sequential decision-making. This novel model estimates the reward function flexibly and non-parametrically, capturing a broad range of function behaviors.

Then, we define some important symbols to construct the GP-CMAB model. Let $h_m(j)$ denote a specific function for each PoI m , which records the expected reward brought by selecting PoI m when it has not been visited for j rounds. In each round r , we use $J_{m,r}$ to denote the number of rounds that have passed since the last data collection at PoI m . As described above, $J_{m,r}$ is employed as a covariate to elucidate how the dynamic reward function h_m changes across different rounds, i.e., $h_m(J_{m,r}) = |D_r^m| \cdot H_m - \lambda(\sigma_m^2 + G_m^2)$. When data on PoI m is collected by UAVs in round r , $J_{m,r}$ resets to 0; Otherwise, it increases by 1. Considering the limited storage space at each PoI, the reward for PoI m cannot increase indefinitely, so we set a threshold J_0 to restrict the value of $J_{m,r}$. Hence, when UAVs visit to the PoI set $\mathcal{P}_r$ in round r , the value of $J_{m,r+1}$ is updated using the following equations:

$$J_{m,r+1} = \begin{cases} 0 & \text{if } m \in \mathcal{P}_r, \\ \min\{J_0, J_{m,r} + 1\} & \text{otherwise.} \end{cases} \quad (17)$$

Next, we estimate the unknown and dynamic reward function $h_m(j)$ under the GP-CMAB model, i.e., utilizing past

observations with the value of $J_{m,r}$ to conduct the reward estimation of PoI m in the r -th round. Specifically, a Gaussian process for each PoI m can be described as a distribution over function spaces, which is characterized by its mean function (denoted by $\mu_m(j) = \mathbb{E}[h_m(j)]$) and a kernel function (denoted by $g_m(j, j') = \mathbb{E}[(h_m(j) - \mu_m(j))(h_m(j') - \mu_m(j'))]$). Let $Y = h_m(j) + \kappa$ be the observed reward for $h_m(j)$, where κ is an i.i.d. Gaussian noise. If we place a Gaussian process prior on $h_m(j)$ and obtain C observations $\mathbf{Y}_{m,C} = (Y_{m,1}, \dots, Y_{m,C})^T$ for PoI m , then we denote the posterior distribution after C observations as $\mathcal{GP}(\mu_m(j; C), g_m(j, j'; C))$. Under the GP-CMAB model, we establish a relationship between these C observations and other points to be explored, enabling non-parametric modeling of the unknown reward function. To be specific, for $\mathbf{g}_C(j) = (g_m(j_1, j), \dots, g_m(j_C, j))^T$ and the semi-positive definite kernel matrix $\mathbf{G}_C = [g_m(j_i, j_l)]_{i,l=1}^C$, the posterior mean and variance based on C observations are:

$$\mu_m(j; C) = \mathbf{g}_C(j)^T (\mathbf{G}_C + \sigma_m^2 \mathbf{I})^{-1} \mathbf{Y}_{m,C}, \quad (18)$$

$$g_m(j, j'; C) = g_m(j, j') - \mathbf{g}_C(j)^T (\mathbf{G}_C + \sigma_m^2 \mathbf{I})^{-1} \mathbf{g}_C(j'). \quad (19)$$

Now, the posterior distribution $\mathcal{GP}$ of the reward function $h_m(j)$ can be expressed as a normal distribution $\mathcal{N}(\mu_m(j; C), \delta_m^2(j; C))$, where $\delta_m^2(j; C) = g_m(j, j; C)$. Let $C_m(r)$ be the number of times PoI m has been selected up to round r . For simplicity, we abbreviate the posterior mean and variance as $\mu_r(m) = \mu_m(J_{m,r}; C_m(r-1))$ and $\delta_r^2(m) = \delta_m^2(J_{m,r}; C_m(r-1))$. In Eqs. (18) and (19), the kernel matrix $\mathbf{G}_C = [g_m(j_i, j_l)]_{i,l=1}^C$ determines the correlation between different points in the input space, eliminating the need for prior assumptions about parameters. The GP-CMAB model subtly leverages these correlations to construct the posterior distribution of the function based on observed data.

While learning the reward functions, we should make decisions about PoI selection based on the learned reward functions. Given λ , we always make instantaneous decisions and greedily choose the arms with higher expected rewards in each round. We define $\mathcal{L}(\mathcal{P}_r, \lambda) = \sum_{m \in \mathcal{P}_r} (|D_r^m| \cdot H_m - \lambda(\sigma_m^2 + G_m^2))$, then the objective of the greedy algorithm is to maximize the expected $\mathcal{L}(\mathcal{P}_r, \lambda)$ in each round for the given λ . To handle the trade-off between exploitation (i.e., selecting the set of PoIs with the highest estimated reward based on learned results) and exploration (i.e., trying some suboptimal PoIs to potentially discover the optimal PoIs), we introduce the concept of Upper Confidence Bound (UCB) into our GP-CMAB model. In detail, we first use Eqs. (18) and (19) to estimate the posterior mean $\mu_r(m)$ and variance $\delta_r^2(m)$. Next, we design a UCB value $\Psi_r(m)$ that accounts for uncertainty in the GP-CMAB model, described as follows:

$$\Psi_r(m) = \mu_r(m) + \alpha_r \delta_r(m), \quad (20)$$

where $\alpha_r = \sqrt{2 \log(M_0 J_0 r^2)}$ is a parameter that implicitly balances the exploration and exploitation trade-off. Here, GP-CMAB enables us to estimate the reward and uncertainty at each PoI accurately. We always select the top M_0 PoIs by sorting the values $\{\Psi_m | m \in \mathcal{M}\}$ in each round. The specific algorithm is detailed in Algorithm 1.

For a fixed value of λ , we prefer to select a set of PoIs with

Algorithm 2: Training process of OFL-UD²

Input: $\mathcal{M}, \mathcal{N}, \mathcal{S}, \mathcal{R}, M_0, \mathbf{w}_0, \eta_r, \epsilon, E_0$, and λ ;

Output: the global model, $\mathbf{w}_R$;

```

1 for Each round  $r = 1, 2, \dots, R$  do
2   The service provider selects a set of PoIs for UAVs
   according to Algorithm 1;
3    $\mathcal{A}_r \leftarrow \text{SModel}(\mathcal{A}_{r-1}, \mathcal{P}_r)$ ;
4   The service provider sends the current global model
    $\mathbf{w}_{r-1}$  and decision variables  $\mathcal{A}_r$  to each UAV;
5   for Each UAV  $n = 1, 2, \dots, N$  in parallel do
6     for Each PoI  $m \in \mathcal{A}_n(r)$  do
7       Travel and collect data at PoI  $m$ ;
8       Get the observation values of  $|D_r^m|, H_m$ ;
9       Estimate  $G_m^2 = \mathbb{E}[\|\nabla F_{r,m}(\mathbf{w}_{r-1}, \xi)\|^2]$  and
        $\sigma_m^2 = \mathbb{E}[\|\nabla F_{r,m}(\mathbf{w}_{r-1}, \xi_m) - \nabla F_{r,m}(\mathbf{w}_{r-1})\|^2]$ ;
10      Calculate the observed reward based on Eq.
       (16), i.e.,  $|D_r^m| \cdot H_m - \lambda(\sigma_m^2 + G_m^2)$ ;
11      Initialize  $\mathbf{w}_{r,n}^0 = \mathbf{w}_{r-1}$ ;
12      for Each local iteration  $k = 0, 1, \dots, K-1$  do
13         $\mathbf{w}_{r,n}^{k+1} = \mathbf{w}_{r,n}^k - \eta_r \nabla F_{r,n}(\mathbf{w}_{r,n}^k, \xi)$ ;
14      Travel back to the ABS according to  $\mathcal{A}_r$ ;
15      Upload the final model parameters  $\mathbf{w}_{r,n}^K$  and re-
       ward observations to the service provider;
16   The service provider aggregates the local models to
   form the global model:  $\mathbf{w}_r = \frac{1}{N} \sum_{i \in \mathcal{N}} p_i \mathbf{w}_{r,i}^K$ ;
17   The service provider learns the dynamic reward func-
   tion for each PoI in the GP-CMAB model;
```

larger UCB values $\Psi_r(m)$ in the t -th round. As previously mentioned, the objective of the PoI selection problem is to solve the Lagrange dual function, i.e., $\min_{\lambda} \max_{\mathcal{P}} \mathcal{L}(\mathcal{P}, \lambda)$. GP-CMAB provides a solution to $\max_{\mathcal{P}} \mathcal{L}(\mathcal{P}, \lambda)$. Next, we need to concentrate on solving $\min_{\lambda} \mathcal{L}(\mathcal{P}^*, \lambda)$, i.e., finding the optimal λ that minimizes $\mathcal{L}(\mathcal{P}^*, \lambda)$. In order to achieve this goal, we treat λ as a hyperparameter and select the optimal λ through grid-search tuning.

C. Path Planning

Through our proposed GP-CMAB-based PoI selection algorithm 1, we have learned the reward function of each PoI over time and selected the PoIs with the highest expected rewards in each round. However, this approach does not fully resolve Problem **P2**, as the complete decision $\mathcal{A}$ must also meet the energy constraints in Eq. (9). In our setup, the server possesses global knowledge of PoI topology and maintains records of the distances between each PoI. Given the set of PoIs $\mathcal{P}_r$ identified by the algorithm 1 and the positions of any UAVs from the last round, we need to schedule N UAVs to minimize the total energy $\sum_{n \in \mathcal{N}} E_n(r)$ required to visit all PoIs in $\mathcal{P}_r$. This scenario presents a typical open-path multi-traveling salesman problem, where multiple UAVs start from their respective ABSs, need to traverse all targets, and only have to reach any ABSs in the end, without returning to their

origin. This is an NP-hard problem. As path planning is not the primary focus of this paper, we use the SModel method from [36] for UAV path planning. The overall decision variables $\mathcal{A}_r$ are then obtained.

D. Framework Design and Analysis

We illustrate the proposed OFL-UD² framework pseudocode in Algorithm 2, which mainly consists of two components: UAV scheduling in unknown environments and federated training. In unknown environments, we need to learn the reward function of unknown PoIs. Specifically, we record the reward values and the corresponding rounds when visiting PoIs (Lines 7-10). Based on these observations, the service provider learns the reward function with respect to different rounds (Line 17). While exploring, the GP-CMAB model makes PoI selection decisions by taking the dynamics and uncertainty of learned rewards into account (Line 2). PoI selection is integrated with path planning to formulate the final UAV scheduling decision (Line 3). Once the scheduling decision is obtained, multiple UAVs perform FL based on the scheduling decision under the coordination of the service provider (Lines 11-16). Online decision-making and FL are performed alternately, and finally, a global model with good generalization and convergence is obtained.

In the subsequent section, we analyze the theoretical performance of the GP-CMAB-based PoI selection algorithm. Regret is commonly used to measure the performance of an online learning algorithm, which is the gap between the cumulative reward of the algorithm and the optimal reward, i.e.,

$$\mathbb{E}[Reg_R] = \sum_{r=1}^R \mathbb{E}[\mathcal{L}(\mathcal{P}_r^*) - \mathcal{L}(\mathcal{P}_r)], \quad (21)$$

where $\mathcal{P}_r^* = \arg \max_{\mathcal{P}_r} \mathcal{L}(\mathcal{P}_r)$ is obtained under the circumstance that all reward functions are known. We use Bayesian regret, which is the expectation taken over the reward function and the actions. Next, we prove the upper bound of the expected cumulative regret for the proposed algorithm.

Theorem 2. *The expected cumulative regret of the GP-CMAB-based PoI selection strategy satisfies the following inequality:*

$$\mathbb{E}[Reg_R] \leq O(\sqrt{MM_0 R \gamma_R \log(MR J_0)}), \quad (22)$$

where γ_R is the maximum information gain after R rounds.

In Theorem 2, we prove that Algorithm 1 achieves the sublinear regret performance. As regret represents the cumulative gap between rewards from our algorithm and the optimal solution, the sublinear regret implies that our algorithm rapidly approaches the optimal solution. In other words, the parameter estimation in Algorithm 1 can converge within limited rounds.

VI. EXPERIMENTAL EVALUATION

This section presents the experimental results. Our primary goal is to optimize UAV decision-making to improve model performance in terms of convergence and generalization. Concretely, we evaluate OFL-UD² mainly from an FL perspective, using test accuracy as the primary metric and comparing it with relevant baselines. We first report the overall performance of the framework and baselines, followed by ablation studies

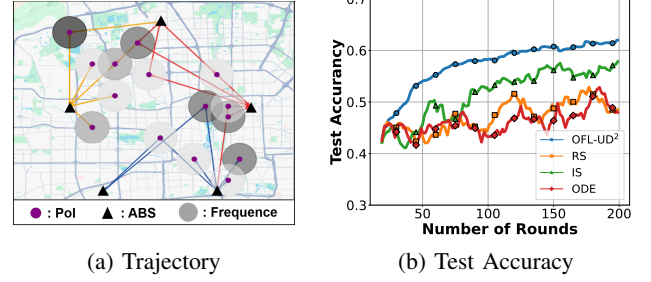


Fig. 3: Test on the Outdoor-images dataset

to assess robustness under varying environmental conditions. The simulations are conducted on an AMAX deep learning workstation with an Intel Xeon Platinum 8358P CPU, an NVIDIA A40 (48GB) GPU, and 80GB RAM.

A. Experimental Setup

Datasets and Models: To demonstrate the performance of OFL-UD² on unknown distributed data, we adopt a real-world dataset collected via crowdsensing and four common datasets: outdoor-images dataset [37], CIFAR-10 [38], SVHN [39], HAM [40], and Animals [41]. The outdoor-images dataset contains a total of 31,601 outdoor environment images, captured in real-time by devices deployed across different regions of Beijing. Each image is labeled with an air pollution level (ranging from 1 to 6) recorded by monitoring stations within the corresponding region. A lightweight ResNet-8 network with 3 residual blocks is employed to infer air quality from these images directly. The CIFAR-10 dataset consists of 60,000 color images, each with a size of $3 \times 32 \times 32$. We use an 8-layer AlexNet [42] with a size of 136MB for CIFAR-10. The SVHN dataset contains 600,000 color images of house numbers, gathered from Google Street View. Given the simplicity of SVHN, we use a custom CNN with two 3×3 convolutional layers for training. The HAM dataset is an image dataset used for skin lesion classification in the medical field. We train a plain CNN model with three 5×5 convolutional layers to process the HAM dataset. Finally, to demonstrate the application of UAVs in wildlife biodiversity monitoring via deep learning [1], we use an animal dataset consisting of 28,000 medium-quality real images of 10 different animal species. For this classification task, we apply the standard ResNet-18 network.

Hardware and Statistical Parameter Settings: In our simulation, after referring to the specifications of commercial drones like Matrice 350 RTK [43], we set $p_1 = 79.86\text{watt}$, $p_2 = 88.63\text{watt}$, $p_3 = 0.02\text{watt}$, $U_p = 120\text{m/s}$, $v_0 = 4.03\text{m/s}$, and $v = 8\text{m/s}$ to calculate the UAV's flight energy consumption. For the energy consumption during the learning phase, we set $e_1 = 0.01\text{J}$ and $e_2 = 80\text{J}$ [18]. To better align with real-world data collection scenarios, we design fine-grained PoI data models from three perspectives: data distribution, data generation patterns, and data quality. In the experiments, training samples at each PoI are independently sampled according to the vector q_m . To simulate non-IID

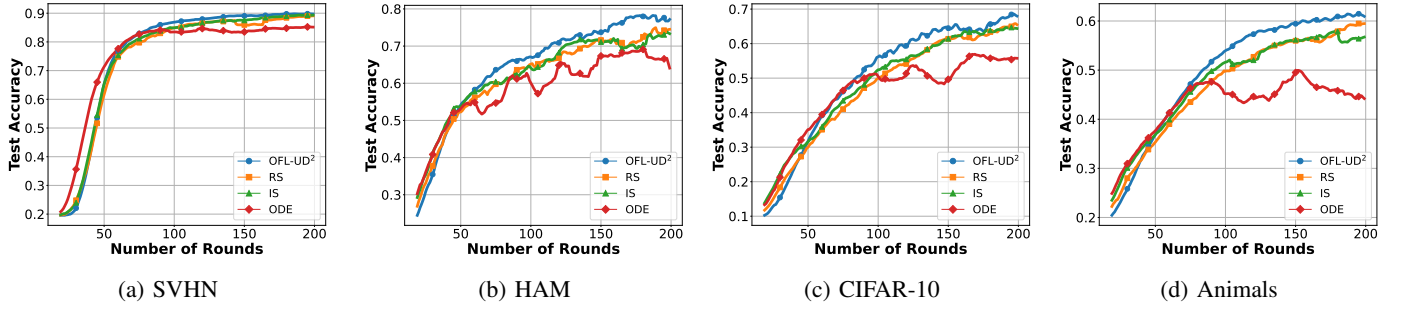


Fig. 4: Test accuracy of four approaches on the four IID datasets

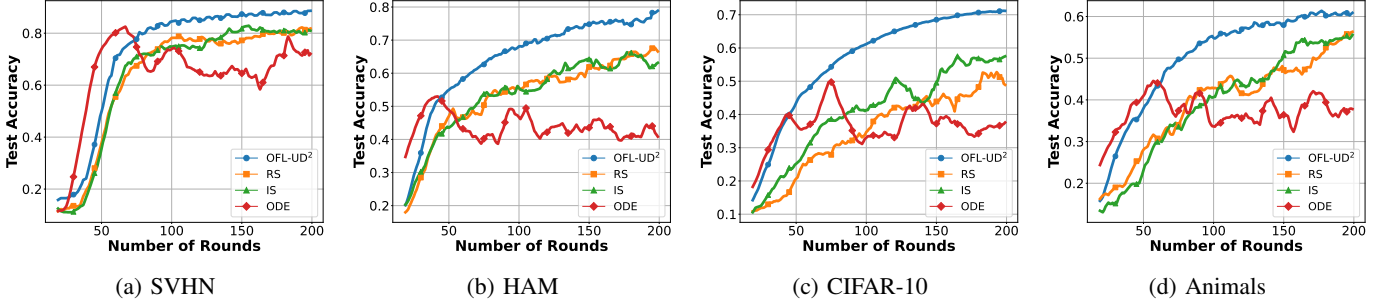


Fig. 5: Test accuracy of four approaches on the four non-IID datasets

datasets, we model the data distribution using the Dirichlet distribution [22], i.e., $q_m \sim \text{Dir}(\beta q_0)$, where $\beta \geq 0$ is a concentration parameter controlling statistical heterogeneity, and q_0 represents a prior label distribution. When $\beta \rightarrow \infty$, all PoIs have a relatively uniform prior distribution (i.e., IID). As $\beta \rightarrow 0$, each PoI holds data samples from only a single class. We specify several values (e.g., $\infty, 1, 0.5, 0.25, 0.2, 0.1$) for β to generate different data distributions, covering a spectrum of heterogeneity. Let $\varsigma = \frac{1}{\beta}$ (i.e., $\varsigma = 0, 1, 2, 4, 5, 10$) denote the Non-IID level. Additionally, we introduce random Gaussian noise of different levels into the data samples to capture differences in data quality across PoIs. Considering that PoIs generate data following different patterns such as event-driven, periodic sampling, or Poisson processes [44], we apply several functions—linear, exponential, and logarithmic models—to reflect the dynamics of data volume. These approaches comprehensively characterize the variability of PoI data in real-world scenarios.

Baselines: Since existing works do not consider applying FL to scenarios where UAVs collect data under unknown environments, they cannot be directly applied to our system. We modify some data selection strategies as baselines. (i) Importance Sampling (IS) [13], [14], [45]: This method uses two indicators: higher loss or higher gradient norm, to reflect the informativeness of the client data and its impact on the global model. Here, we periodically compute the gradient norms at each PoI on the model and prioritize PoIs with higher gradient norms. (ii) Online Data Selection (ODE) [15]: It is a state-of-the-art online data selection method on devices, which uses the projection of local gradients onto the global gradient as a value metric for data selection. (iii) Random Sampling (RS): UAVs randomly select PoIs to collect data in each round.

B. Overall Performance

First, we evaluate the performance of OFL-UD² on the real-world distributed outdoor-images dataset using 3 UAVs. The resulting trajectory diagram is shown in Fig. 3a. Each colored line represents the flight trajectory of a UAV, and darker-colored PoIs indicate a higher frequency of visits. We can observe that some PoIs have obvious priorities when being visited. Meanwhile, Fig. 3b demonstrates that the test accuracy of OFL-UD² after 200 rounds is 62.01%, which is about 4.18%, 13.49%, and 14.04% higher than those of IS, ODE, and RS, respectively. These findings confirm that OFL-UD² effectively selects PoIs to maximize model training performance in unknown and unpredictable environments.

Second, we conduct a set of experiments on four widely used FL datasets under the IID setting. In the data partitioning process, different PoIs are configured with identical distributions and noise but varying data capacities. The training processes of these methods are illustrated in Fig. 4. OFL-UD² indeed exhibits a slight advantage over the baselines. For example, as depicted in Fig. 4c, OFL-UD² separately improves the final test accuracy by about 2.50%, 3.61%, and 12.23% for AlexNet on CIFAR-10, compared to RS, IS, and ODE. Nevertheless, OFL-UD² and the baselines exhibit similar performance for CNN on SVHN. This is because all algorithms differ only in the data volume under the IID setting, allowing OFL-UD² to prioritize PoIs with larger data quantities. Since the features in the SVHN dataset are relatively simple, increasing the data volume will not be conducive to model training. As a result, OFL-UD² does not demonstrate a significant advantage over the baselines on this dataset.

Third, we manually adjust four FL datasets to simulate more complex heterogeneity representative of real-world scenarios.

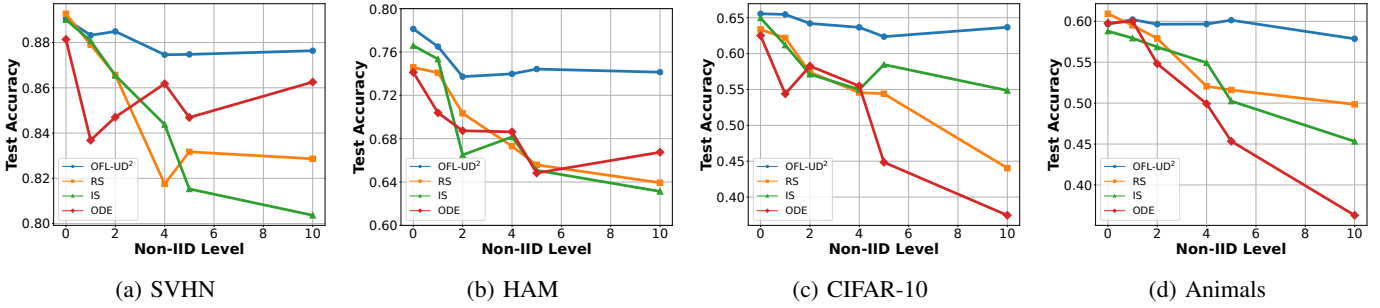


Fig. 6: Test accuracy varies with different non-IID levels.

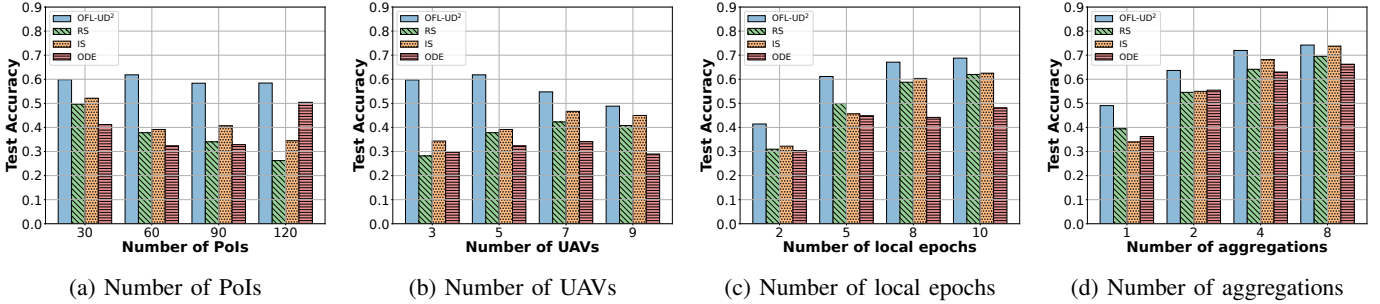


Fig. 7: Effects of four key arguments.

We set the non-IID level $\varsigma = 10$ across 45 PoIs, with data uniformly distributed across 15 PoIs to emphasize inter-PoI variation. We observe that OFL-UD² achieves the highest accuracy and faster convergence among these methods in Fig. 5. For instance, as shown in Fig. 5a, OFL-UD² improves the test accuracy by about 6.83%, 7.68%, and 16.58% for CNN on SVHN over 200 training rounds, compared to RS, IS, and ODE, respectively. Additionally, as illustrated in Fig. 5b, OFL-UD² can separately speed up training to around 50% accuracy by about 1.2 $\times$, 1.9 $\times$, and 3.0 $\times$, compared to ODE, IS, and RS. It is noteworthy that ODE shows faster convergence rates in the early stages on some datasets. This is because bandit-based algorithms typically require an initial exploration phase to gather information for initialization. Moreover, by comparing Figs. 4 and 5, we observe that the performance gap is less pronounced in the IID setting compared to the non-IID setting. This indicates that OFL-UD² performs better under heterogeneous data distributions and noise. It also indirectly suggests that, when evaluating the utility of PoI data, the impact of data distribution and sample quality on the training process is more significant than data volume alone.

C. Ablation Study

Effect of Non-IID Levels: We evaluate the test accuracy of these schemes over four datasets under different Non-IID levels within a given number of 100 completion rounds. As shown in Fig. 6, where the horizontal axis marks the Non-IID levels of the datasets, the test accuracy of model training across all four methods decreases as the Non-IID level increases. For example, by Fig. 6c, when training AlexNet on CIFAR-10 with a Non-IID level of 10, OFL-UD² achieves an accuracy of 63.67%, which is 11.58%, 19.63%, and 27.83% higher than IS, RS, and ODE, respectively. OFL-UD² investigates the benefits

of using GP-CMAB to dynamically estimate and select PoIs with higher reward, ensuring that better PoIs are always chosen in changing scenarios. Compared to data utility evaluation methods like ODE and IS, OFL-UD² tends to select data with a more uniform distribution, mitigating data heterogeneity among different clients. Additionally, by selecting samples with smaller gradient variance, OFL-UD² can also accelerate the convergence speed of the online FL.

Effect of Different Choice of Parameters: In order to verify the parameter choices and confirm their effects, we tune four key parameters in OFL-UD², including the number of PoIs, the number of UAVs, the number of local iterations, and the number of aggregations in one round. The number of PoIs reflects the scale of data collection required by UAVs. By Fig. 7a, with a fixed number of UAVs, the final performance of RS shows a significant decline as the number of PoIs to be collected increases. In contrast, OFL-UD² demonstrates a certain level of robustness. This is because, after completing the exploration phase, OFL-UD² tends to select a set of PoIs with higher data utility, resulting in more stable performance. According to Fig. 7b, we set the optimal number of UAVs as 5 when the number of PoIs is 60. The numbers of local iterations and aggregations are classic parameters in FL, reflecting the degree of data training at each PoI in each round. We observe that the test accuracy of all methods gradually improves with the increase of local epochs and aggregations.

Effect of Key Strategies: Our OFL-UD² incorporates two key designs to enhance performance: selecting PoIs based on gradient convergence and using GP for dynamic reward estimation. We conduct experiments with AlexNet on CIFAR-10 under IID and non-IID distributions. In the IID setting, noise and distributions across PoIs are uniform. In the non-IID setting, PoIs have heterogeneous noise and distributions. We

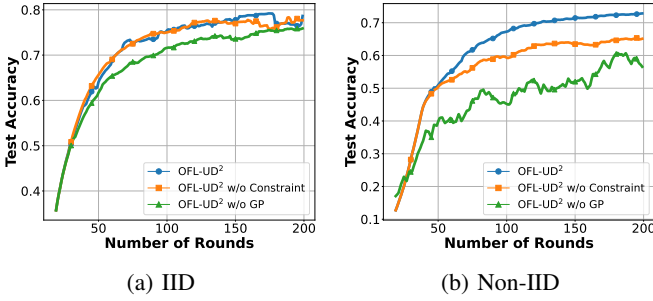


Fig. 8: Effects of convergence constraint and GP

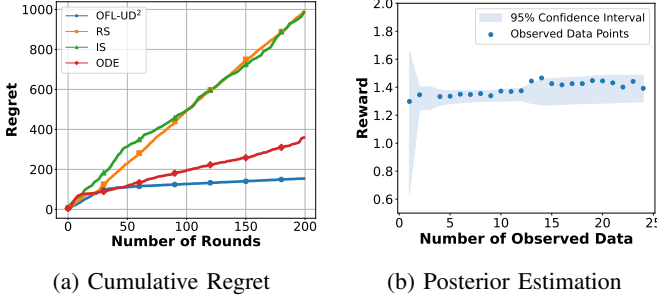


Fig. 9: Efficiency of GP-CMAB

use the OFL-UD² without convergence constraint (OFL-UD² w/o Constraint) and OFL-UD² without GP (OFL-UD² w/o GP) as baselines. Specifically, in OFL-UD² w/o Constraint, UAVs make decisions based only on PoI utility estimation by GP-CMAB, without factoring in PoIs with better convergence. While in (OFL-UD² w/o GP), UAVs learn static reward values and only choose to select the same PoI after fixed rounds. By Fig. 8, OFL-UD² w/o Constraint converges similarly as OFL-UD² because of the IID setting. Under the non-IID setting, OFL-UD² improves test accuracy by approximately 7.74% and 16.28% over OFL-UD² w/o Constraint and OFL-UD² w/o GP, respectively. This highlights the performance gains enabled by these two key designs.

Efficiency of GP-CMAB: In online learning systems, cumulative regret is used to evaluate an algorithm's learning efficiency, convergence speed, and adaptability. To assess the efficiency of our algorithm in unknown environments, we test its cumulative regret compared to the optimal solution for ResNet-8 on the outdoor-images dataset, as shown in Fig. 9a. We observe that the regret becomes stable after approximately 50 rounds. It indicates that the GP-CMAB-based PoI selection strategy can rapidly obtain expected parameter estimation and converge to the optimal solution within limited rounds. This further validates its ability to make effective decisions in unknown real-world scenarios. Fig. 9b shows the impact of observed data variability on parameter estimates. GP-CMAB can adapt to environmental uncertainty by dynamically and adaptively adjusting the estimated confidence interval.

VII. CONCLUSION

In this paper, we explicitly consider a novel scenario where UAVs collect data by dynamically visiting PoIs to perform an FL task. The complex relationship between UAV decision-

making and the performance of the trained model has not been explored in existing works. We then present the design, implementation, and evaluation of OFL-UD², which is an online FL framework on unknown distributed data, consisting of two components: GP-CMAB-based PoI selection and path planning. Our analysis shows that OFL-UD² can maximize the generalization of the model while ensuring convergence and achieve a sublinear rate of cumulative regret compared to the optimal decision. Empirical results demonstrate that OFL-UD² significantly outperforms existing data selection baselines in unknown dynamic environments on various datasets.

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APPENDIX

A. Proof of Theorem 1

Following the conventions established in previous studies [46], we analyze the difference between the expected global loss $\mathbb{E}[F(\mathbf{w}_r)]$ and the optimal value $F(\mathbf{w}^*)$ by analyzing the difference between the models $\mathbf{w}_r$ and $\mathbf{w}^*$ in each round. We define $\Omega_r = \|\mathbf{w}_r - \mathbf{w}^*\|^2$, then

$$\begin{aligned} \Omega_r &= \Omega_{r-1} + \mathbb{E} \left\| \sum_{i \in \mathcal{P}_r} \frac{1}{M_0} (\mathbf{w}_r^i - \mathbf{w}_{r-1}) \right\|^2 \\ &\quad + 2\mathbb{E} \langle \mathbf{w}_{r-1} - \mathbf{w}^*, \sum_{i \in \mathcal{P}_r} \frac{1}{M_0} (\mathbf{w}_r^i - \mathbf{w}_{r-1}) \rangle. \end{aligned} \quad (23)$$

For simplicity, we denote $A_1 = \mathbb{E} \left\| \sum_{i \in \mathcal{P}_r} \frac{1}{M_0} (\mathbf{w}_r^i - \mathbf{w}_{r-1}) \right\|^2$ and $A_2 = \mathbb{E} \langle \mathbf{w}_{r-1} - \mathbf{w}^*, \sum_{i \in \mathcal{P}_r} \frac{1}{M_0} (\mathbf{w}_r^i - \mathbf{w}_{r-1}) \rangle$. By Cauchy-Schwarz inequality, we have

$$\begin{aligned} A_1 &= \left\| -\sum_{m \in \mathcal{P}_r} \frac{\eta_r}{M_0} \sum_{k=0}^{K-1} \nabla F_{r,m}(\mathbf{w}_r^{m,k}; \xi_r^m) \right\|^2 \\ &\leq \frac{K\eta_r^2}{M_0} \sum_{m \in \mathcal{P}_r} \sum_{k=0}^{K-1} \|\nabla F_{r,m}(\mathbf{w}_r^{m,k}; \xi_r^m)\|^2. \end{aligned} \quad (24)$$

According to Assumptions 3 and 4, we have

$$\begin{aligned} &\|\nabla F_{r,m}(\mathbf{w}_r^{m,k}; \xi_r^m)\|^2 \\ &= \|\nabla F_{r,m}(\mathbf{w}_r^{m,k}; \xi_r^m) - \nabla F_{r,m}(\mathbf{w}_r^{m,k})\|^2 + \|\nabla F_{r,m}(\mathbf{w}_r^{m,k})\|^2 + \\ &\quad 2\mathbb{E} \langle \nabla F_{r,m}(\mathbf{w}_r^{m,k}; \xi_r^m) - \nabla F_{r,m}(\mathbf{w}_r^{m,k}), \nabla F_{r,m}(\mathbf{w}_r^{m,k}) \rangle \\ &\leq G_m^2 + \sigma_m^2. \end{aligned} \quad (25)$$

For A_2 , we divide it into two terms. Specifically,

$$\begin{aligned} A_2 &= \mathbb{E} \langle \mathbf{w}_{r-1} - \mathbf{w}^*, -\sum_{m \in \mathcal{P}_r} \frac{\eta_r}{M_0} \nabla F_{r,m}(\mathbf{w}_r^{m,k}; \xi_r^m) \rangle + \\ &\quad \mathbb{E} \langle \mathbf{w}_{r-1} - \mathbf{w}^*, -\sum_{m \in \mathcal{P}_r} \frac{\eta_r}{M_0} \sum_{k=1}^{K-1} \nabla F_{r,m}(\mathbf{w}_r^{m,k}; \xi_r^m) \rangle. \end{aligned} \quad (26)$$

Let $B_1 = \mathbb{E} \langle \mathbf{w}_{r-1} - \mathbf{w}^*, -\sum_{m \in \mathcal{P}_r} \frac{\eta_r}{M_0} \nabla F_{r,m}(\mathbf{w}_r^{m,k}; \xi_r^m) \rangle$ and $B_2 = \mathbb{E} \langle \mathbf{w}_{r-1} - \mathbf{w}^*, -\sum_{m \in \mathcal{P}_r} \frac{\eta_r}{M_0} \sum_{k=1}^{K-1} \nabla F_{r,m}(\mathbf{w}_r^{m,k}; \xi_r^m) \rangle$. Then, we aim to bound B_1 and B_2 respectively. Given the assumption that $F_{r,m}(\cdot)$ is ν -strongly convex and the fact that $F_{r,m}^* \leq F_{r,i}(\mathbf{w}_{r-1})$, it follows that

$$\begin{aligned}
B_1 &= -\sum_{m \in \mathcal{P}_r} \frac{\eta_r}{M_0} \langle \mathbf{w}_{r-1} - \mathbf{w}^*, \nabla F_{r,m}(\mathbf{w}_{r-1}) \rangle \\
&\leq \sum_{m \in \mathcal{P}_r} \frac{\eta_r}{M_0} (F_{r,m}(\mathbf{w}^*) - F_{r,m}^*) - \frac{\nu \eta_r}{2} \Omega_{r-1} \\
&\leq \sum_{m \in \mathcal{P}_r} \frac{\eta_r}{M_0} \frac{1}{2\nu} \|\nabla F_{r,m}(\mathbf{w}^*)\|^2 - \frac{\nu \eta_r}{2} \Omega_{r-1} \quad (27) \\
&\leq \sum_{m \in \mathcal{P}_r} \frac{\eta_r}{2M_0\nu} G_m^2 - \frac{\nu \eta_r}{2} \Omega_{r-1}. \quad (28)
\end{aligned}$$

Here, the inequality $F_{r,m}(\mathbf{w}) - F_{r,m}(\mathbf{w}^*) \leq \frac{1}{2\nu} \|\nabla F_{r,m}(\mathbf{w})\|^2$ is applied to Eq. (27), and Eq. (28) is obtained according to Assumption 4. For B_2 , by Cauchy-Schwarz inequality and AM-GM inequality, we have

$$B_2 \leq \frac{\nu \eta_r}{4} \Omega_{r-1} + \frac{\eta_r (K-1)^2}{M_0 \nu} \sum_{m \in \mathcal{P}_r} (G_m^2 + \sigma_m^2). \quad (29)$$

By combining Eqs. (23) ~ (29), we get the recursion as

$$\Omega_r \leq (1 - \frac{\nu \eta_r}{2}) \Omega_{r-1} + \frac{\beta_1}{M_0 \nu} \sum_{m \in \mathcal{P}_r} (G_m^2 + \sigma_m^2). \quad (30)$$

Therefore, we can derive the following inequality:

$$\Omega_R \leq (1 - \frac{\nu \tilde{\eta}}{2})^R \Omega_0 + \frac{\beta_1}{M_0 \nu} \sum_{r=1}^R \sum_{m \in \mathcal{P}_r} (G_m^2 + \sigma_m^2), \quad (31)$$

where $\tilde{\eta} = \max_r \{\eta_r\}$, and $\beta_1 = \tilde{\eta}[\nu \tilde{\eta} K^2 + 2(K-1)^2 + 1]$. Finally, by the L -smoothness of $F(\cdot)$, we have

$$\begin{aligned}
\mathbb{E}[F(\mathbf{w}_R)] - F(\mathbf{w}^*) &\leq \frac{L}{2} (1 - \frac{\nu \tilde{\eta}}{2})^R \|\mathbf{w}_0 - \mathbf{w}^*\|^2 \\
&\quad + \frac{L \beta_1}{2M_0 \nu} \sum_{r=1}^R \sum_{m \in \mathcal{P}_r} [G_m^2 + \sigma_m^2]. \quad (32)
\end{aligned}$$

The proof of the Theorem 1 is now completed.

B. Proof of Theorem 2

Before proving the Theorem 2, we present three lemmas.

Lemma 1. (Deviation estimation) For any PoI m and constant $a \leq 0$ at episode r , it holds that

$$\begin{aligned}
&\int_a^\infty \mathbb{P}(h_m(j) - \mu_r(m) \geq x | \mathcal{F}_{r-1}) dx \\
&\leq \sqrt{2\pi} \delta_r(m) \exp \left\{ -\frac{a^2}{2\delta_r^2(m)} \right\}. \quad (33)
\end{aligned}$$

Proof. By utilizing the normality of the posterior $h_m(r)$, i.e., $h_m(r) \sim \mathcal{N}(\mu_r(m), \delta_r(m))$, we have

$$\begin{aligned}
&\int_a^\infty \mathbb{P}(h_m(j) - \eta_r(m) \geq x | \mathcal{F}_{r-1}) dx \\
&\leq \int_a^\infty \exp \left\{ -\frac{x^2}{2\delta_r^2(m)} \right\} dx \\
&= \sqrt{2\pi} \delta_r(m) \int_a^\infty \frac{1}{\sqrt{2\pi} \delta_r(m)} \exp \left\{ -\frac{x^2}{2\delta_r^2(m)} \right\} dx \\
&\leq \sqrt{2\pi} \delta_r(m) \exp \left\{ -\frac{a^2}{2\delta_r^2(m)} \right\}.
\end{aligned}$$

The proof of the lemma is now completed. $\square$

Lemma 2. (Instantaneous regret) Assume we select $\mathcal{P}_r$ which have the highest upper confidence bound Ψ at time r , then instantaneous regret w.r.t. $\mathcal{P}_r$ satisfies

$$\mathbb{E} \left[\sum_{i \in \mathcal{P}_r^*} h_i(j) - \sum_{i \in \mathcal{P}_r} h_i(j) | \mathcal{F}_{r-1} \right] \leq \frac{\sqrt{2\pi}}{r^2} + \alpha_r \sum_{i \in \mathcal{P}_r} \delta_r(i). \quad (34)$$

Proof. We first decompose the regret as follows.

$$\begin{aligned}
&\mathbb{E} \left[\sum_{i \in \mathcal{P}_r^*} h_i(j) - \sum_{i \in \mathcal{P}_r} h_i(j) | \mathcal{F}_{r-1} \right] \\
&\leq \mathbb{E} \left[\sum_{i \in \mathcal{P}_r^*} [h_i(j) - (\mu_r(i) + \alpha_r \delta_r(i))] \right. \\
&\quad \left. + \sum_{i \in \mathcal{P}_r} [(\mu_r(i) + \alpha_r \delta_r(i)) - h_i(j)] | \mathcal{F}_{r-1} \right] \quad (35) \\
&= \sum_{i \in \mathcal{P}_r^*} \mathbb{E}[h_i(j) - \mu_r(i) - \alpha_r \delta_r(i) | \mathcal{F}_{r-1}] \\
&\quad + \alpha_r \sum_{i \in \mathcal{P}_r} \delta_r(i). \quad (36)
\end{aligned}$$

Eq. (35) holds because we select the set $\mathcal{P}_r$ with the largest Ψ values. Eq. (36) holds because $\mu_r(i) = \mathbb{E}[h_i(j) | \mathcal{F}_{r-1}]$. Based on the fact that for any random variable X , it holds $\mathbb{E}[X] \leq \mathbb{E}[X \cdot \mathbb{1}_{\{X > 0\}}] = \int_0^\infty \mathbb{P}(X \geq x) dx$, we bound the first term in Eq. (36) as follows:

$$\begin{aligned}
&\sum_{i \in \mathcal{P}_r^*} \mathbb{E}[h_i(j) - \mu_r(i) - \alpha_r \delta_r(i) | \mathcal{F}_{r-1}] \\
&\leq \sum_{i \in \mathcal{M}} \sum_{j \leq J_0} \sqrt{2\pi} \delta_r(i) \exp \left\{ -\frac{\alpha_r^2 \delta_r^2(i)}{2\delta_r^2(i)} \right\} \quad (37)
\end{aligned}$$

$$\leq \sum_{i \in \mathcal{M}} \sum_{j \leq J_0} \frac{\sqrt{2\pi}}{M J_0 r^2} = \frac{\sqrt{2\pi}}{r^2}. \quad (38)$$

In Eq. (37), we utilize the result from Lemma 1. Eq. (38) holds due to the fact that $\delta_r^2(m) \leq 1$. $\square$

Lemma 3. (Bounding GP variance)

$$\sum_{r=1}^R \sum_{i \in \mathcal{P}_r} \delta_r^2(i) \leq C_1 M \gamma_R, \quad (39)$$

where $C_1 = 1/(\log(1 + \delta^{-2}))$.

Proof. Utilizing the fact that the maximal information gain is increasing in the number of data points and the results of Lemma 5.4 in [47], we have

$$\begin{aligned}
\sum_{r=1}^R \sum_{i \in \mathcal{P}_r} \delta_r^2(i) &= \sum_{i=1}^M \sum_{n=1}^{N_m(r)} \delta_m^2(J_m^{(n)}, n-1) \\
&\leq C_1 \sum_{i=1}^M \gamma_{N_j(R)} \leq C_1 M \gamma_R.
\end{aligned}$$

Now, we have completed the proof of the lemma. $\square$

Then, we come back to the proof of Theorem 2. Using Lemma 1, Lemma 3, and Lemma 2, it follows that

$$\begin{aligned}
\mathbb{E}[Reg_R] &\leq \sum_{r=1}^R \mathbb{E} \left[\sum_{i \in \mathcal{P}_r^*} h_i(j) - \sum_{i \in \mathcal{P}_r} h_i(j) | \mathcal{F}_{r-1} \right] \\
&\leq \sum_{r=1}^R \mathbb{E} \left[\frac{\sqrt{2\pi}}{r^2} + \alpha_r \sum_{i \in \mathcal{P}_r} \delta_r(i) \right] \\
&\leq \sum_{r=1}^R \mathbb{E} \left[\frac{\sqrt{2\pi}}{r^2} + \sqrt{N} \alpha_r \sqrt{\sum_{i \in \mathcal{P}_r} \delta_r^2(i)} \right] \\
&\leq \frac{\sqrt{2\pi}^{\frac{5}{2}}}{6} + \sum_{r=1}^R \sqrt{N} \sqrt{2 \log(M J_0 r^2)} \mathbb{E} \left[\sqrt{\sum_{i \in \mathcal{P}_r} \delta_r^2(i)} \right] \\
&\leq \frac{\sqrt{2\pi}^{\frac{5}{2}}}{6} + \sqrt{4N \log(M J_0 R)} \sum_{r=1}^R \mathbb{E} \left[\sqrt{\sum_{i \in \mathcal{P}_r} \delta_r^2(i)} \right] \\
&\leq \frac{\sqrt{2\pi}^{\frac{5}{2}}}{6} + \sqrt{4NR \log(M J_0 R)} \mathbb{E} \left[\sqrt{\sum_{r=1}^R \sum_{i \in \mathcal{P}_r} \delta_r^2(i)} \right] \\
&\leq \frac{\sqrt{2\pi}^{\frac{5}{2}}}{6} + \sqrt{4MNR \gamma_R \log(M J_0 R)} \sqrt{C_1},
\end{aligned}$$

where $C_1 = 1/(\log(1 + \delta^{-2}))$ and the last inequality follows by Lemma 3.

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